Season creep

For several decades, people have observed leaves and flowers appearing earlier in spring. In the past, these claims were often dismissed as lacking “hard science,” such as facts, figures, or datasets. When Camille Parmesan and Gary Yohe published evidence in 2003—based on an analysis of more than 1,700 species—they demonstrated that change was very real. Their data showed that spring change was indeed taking place earlier—by an average of 2.3 days per decade. Studies by other scientists in recent years have supported their findings.

Many of the changes that take place in plants are governed by temperature, including growth spurts; the appearance of leaves, flowers, and fruit; and leaves dying in fall. Most food chains start with plants, so these changes affect grazers and browsers, from rabbits to deer, and pollinators, including bees and butterflies. All of these are at the bottom of the food chain (primary consumers). If they struggle to find food, those that prey on them (secondary consumers) also suffer from the absence of prey.

Effects of climate change

A warmer Earth produces many effects. In most cooler parts of the world, the frost-free season is longer than before, providing a longer growing season for plants. As some regions become drier and some wetter, bouts of extremely heavy rainfall and flooding have become more common. Toxic algal blooms in lakes are occurring more frequently. Ice cover in polar regions is also decreasing. All these changes have affected and will continue to affect animal and plant behavior.

Since 1993, the European Environmental Agency (EEA) has worked in earnest to pull together data from thousands of studies—dating back to at least 1943—to create a picture of spring creep in Europe. The EEA’s evidence shows earlier dates for plants producing pollen, frogs spawning,